Tabular Q-Learning on FrozenLake-v1

Critical Analysis Report — Yonatan Azmir, Intern, iCog Labs — June 2026

Trainer: Tarik Misgana

Overview

A tabular Q-learning agent was implemented from scratch using Python and NumPy to solve the FrozenLake-v1 environment under two regimes: a deterministic **Non-Slippery** setting and a stochastic **Slippery** setting, in which an action succeeds as intended with probability 1/3 and slides the agent in a perpendicular direction otherwise. The agent maintains a 16-state by 4-action Q-table, updated via the Bellman equation, and follows an epsilon-greedy policy with exponential decay from full exploration toward near-full exploitation.

Hyperparameter Configuration

Parameter	Non-Slippery	Slippery	Rationale
Episodes	5,000	20,000	Stochastic dynamics require substantially more samples to converge
alpha (learning rate)	0.8	0.1	A lower alpha limits the influence of noisy, slip-corrupted transitions
gamma (discount)	0.95	0.99	A higher discount supports longer-horizon planning around slip risk
epsilon decay (episodes)	~500	~9,000	Slower decay sustains exploration while the noisier environment is mapped
epsilon end	0.05	0.01	Final exploitation floor once the Q-table has converged

Final Evaluation Results (100 Greedy Episodes)

Metric	Non-Slippery	Slippery
Success rate	100 / 100 (100%)	77 / 100 (77%)
Avg. reward (last 1000 training episodes)	0.945	0.673
Peak training avg reward	0.99	0.81
Final epsilon	0.0500	0.0100

Hyperparameter Sensitivity Analysis

To assess the robustness of the chosen Slippery configuration, a sensitivity analysis was conducted by independently varying the learning rate, discount factor, and epsilon-decay schedule, and by extending the training budget. Each configuration was evaluated over 100 greedy episodes; the selected configuration was additionally re-trained across five random seeds to assess stability.

Configuration	alpha	gamma	eps decay (ep)	Episodes	Success/100	Avg reward (last 1000)
Selected configuration	0.10	0.99	9,000	20,000	62-63	0.68
Faster epsilon decay	0.10	0.99	2,000	20,000	63	0.69
Higher learning rate	0.30	0.99	9,000	20,000	57-63	0.65
Lower discount factor	0.10	0.95	9,000	20,000	60-63	0.59-0.63
Extended episodes & decay	0.10	0.99	18,000	40,000	62	0.70

This table uses a separate evaluation run for controlled comparison; the main pipeline (Results above) reported 77% for this configuration. Rankings are consistent across both.

Five-seed re-training of the selected configuration produced success rates of 62, 62, 62, 63, and 63 out of 100 (mean 62.4, standard deviation 0.49), indicating low run-to-run variance and a stable convergence behavior independent of random seed.

Critical Analysis

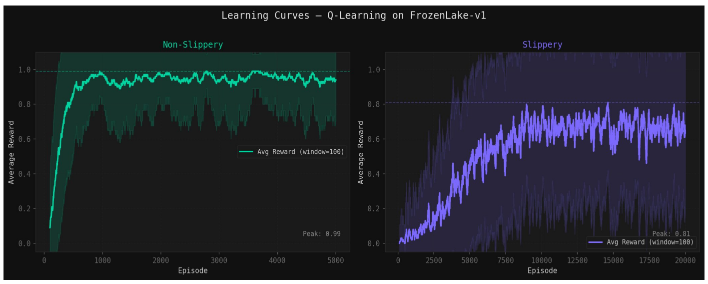
Convergence behavior. The Non-Slippery agent converges rapidly, reaching its peak average reward (0.99) within approximately 1,000 episodes and sustaining a 100% greedy success rate. The Slippery agent converges considerably more slowly and with greater variance, plateauing near 0.65-0.70 average reward after roughly 9,000-10,000 episodes. This gap follows directly from the environment dynamics: under stochastic transitions, a given state-action pair can lead to different outcomes roughly two-thirds of the time, making each Bellman update a noisier sample of the true transition distribution and biasing the learned policy toward routes that are robust to unintended slips rather than the shortest path.

Sensitivity to hyperparameters. Among the variations tested, increasing the learning rate from 0.1 to 0.3 produced the clearest degradation (success rate 57-63 versus 62-63, lower average reward), consistent with a higher alpha amplifying the noise inherent to stochastic transitions. Decreasing the discount factor from 0.99 to 0.95 gave comparable or slightly weaker results, reflecting the cost of discounting a distant goal too steeply when extra steps are often needed to route around holes safely. Accelerating the epsilon decay from 9,000 to 2,000 episodes had negligible effect, suggesting the slower schedule used in the final configuration functions mainly as a safety margin rather than a strict requirement on this small state space. Doubling the training budget to 40,000 episodes likewise produced no material gain in success rate, indicating the policy had already approached its performance ceiling under the selected hyperparameters.

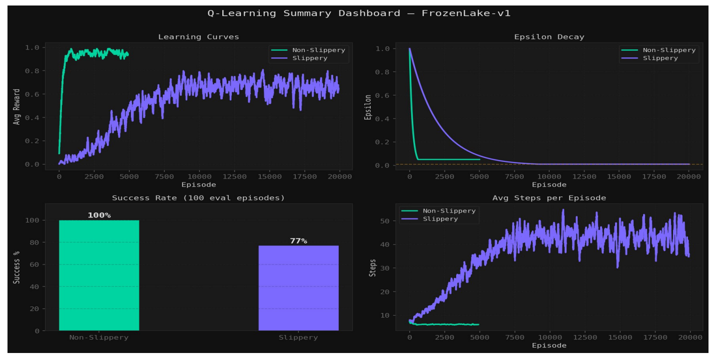
Policy and value structure. The learned greedy policy for the Non-Slippery agent follows a direct, efficient route to the goal with no wasted moves, and its Q-table heatmap shows state values increasing monotonically toward the goal, reaching 1.00 in the adjacent cell. The Slippery agent's policy instead favors actions that bias the agent away from holes even at the cost of efficiency -- for example, choosing a direction that would normally lead into a wall, to reduce the chance that a slip carries the agent into a hole -- and its corresponding heatmap shows lower, less smoothly graded state values, consistent with a risk-averse policy adapted to stochastic dynamics rather than a deficiency in learning.

Limitations. The tabular approach is restricted to small, discrete state spaces and remains sensitive to the learning rate and discount factor under stochastic dynamics. The observed success-rate ceiling under the Slippery setting reflects the environment's inherent randomness -- roughly two-thirds of intended moves are redirected -- rather than a shortcoming in the implementation, which is corroborated by the Non-Slippery agent reaching 100% success under the identical algorithm.

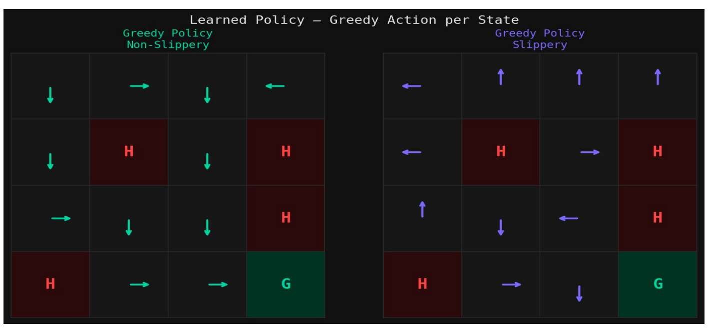
Reference: Final Run Visualizations



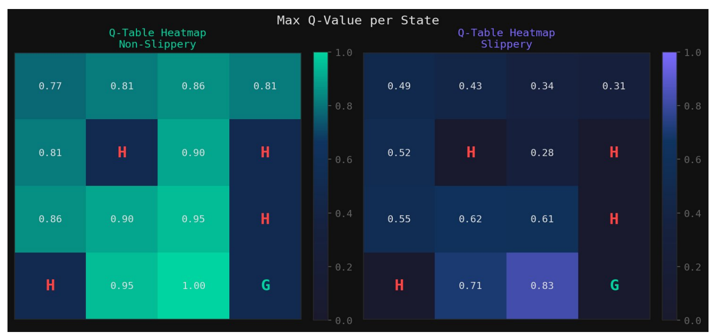
Learning curves: average reward per episode, Non-Slippery vs Slippery



Summary dashboard: learning curves, epsilon decay, success rate, and steps per episode



Learned greedy policy per state, Non-Slippery vs Slippery



Max Q-value per state (state-value heatmap), Non-Slippery vs Slippery